

Assignment: Module 2 | Linear Programming Model |

Link to repository:

<https://github.com/Jdemore/64018-jdemore4>

Direct link to assignment PDF:

https://github.com/Jdemore/64018-jdemore4/blob/main/Assignments/Jdemore4_2.pdf

1. Back Savers

Company: Back Savers { Backpack Manufacturer }

Products: Collegiate (C), Mini (M)

Material in: 5000 ft² of Nylon / week

Limits & Info:

1000C and 1200M sales / week

Labor at 45min per C, 40min per M

Unit profit: 32C, 24M

35 Laborers, 40 hours / week each

a. **Decision Variables:**

C "Collegiate backpacks produced per week"

M "Mini backpacks produced per week"

b. **Objective function:** Maximize profit

Profit(C) = 32

Profit(M) = 24

Profit as "P"

Max P = (32C + 24M)

c. **Constraints** (resource, demand, labor)

Nylon:

$3C + 2M \leq 5000$

Demand:

$C \leq 1000$

$M \leq 1200$

Labor: (35 workers * 40 hours weekly * 60 minutes = 84,000 minutes)

$45C + 40M \leq 84,000$

Technicality: Can't be negative.

$C \geq 0$

$M \geq 0$

d. **Mathematical Formula:**

Max P = 32C + 24M

Given constraints:

Nylon: $3C + 2M \leq 5000$

Collegiate Demand: ≤ 1000

Mini Demand: ≤ 1200

Labor: $45C + 40M \leq 84000$

Non-negativity: { C, M } ≥ 0

In short: Decide how many Collegiate backpacks (C) and Mini backpacks (M) to make, per week.

Goal: Maximize Profit (in \$)

Limits: Nylon supply (in ft²), sales demand (per-backpack), availability of labor (in minutes)

2. Weigelt Corporation

Company: Weigelt Corporation

Products: Small, Medium, Large

Limits & info:

Plant Capacity:	P1, 750	P2, 900	P3, 450
Storage Space:	P1, 13000	P2, 12000	P3, 5000
Storage Used:	S, 12	M, 15	L, 20
Sales:	S, 750	M, 1200	L, 900
Profit:	S, 300	M, 360	L, 420

Each plant must use the same percentage of their excess capacity to produce the new product

a. Decision Variables:

S "Small units"	P₁ "Plant 1"
M "Medium units"	P₂ "Plant 2"
L "Large units"	P₃ "Plant 3"

P₁S, P₁M, P₁L, P₂S, P₂M, P₂L, P₃S, P₃M, P₃L

b. Linear Programming Model

Objective Function:

$$\text{Max Profit} = 300(P_1S + P_2S + P_3S) + 360(P_1M + P_2M + P_3M) + 420(P_1L + P_2L + P_3L)$$

Given Constraints: "P" as Plant

Percentage Used: (P1) = Percentage Used (P2) = Percentage Used (P3)

$$\text{As } (P_1S + P_1M + P_1L)/750 = (P_2S + P_2M + P_2L)/900 = (P_3S + P_3M + P_3L)/450$$

Factor out LCD as 4500

$$P_1/750 = P_2/900 = P_3/450 \implies 6P_1 = 5P_2 = 10P_3$$

So:

$$6P_1 - 5P_2 = 0$$

$$6(P_1S + P_1M + P_1L) - 5(P_2S + P_2M + P_2L) = 0$$

$$5P_2 - 10P_3 = 0 \implies P_2 - 2P_3 = 0$$

$$(P_2S + P_2M + P_2L) - 2(P_3S + P_3M + P_3L) = 0$$

Capacity:

$$P_1S + P_1M + P_1L \leq 750$$

$$P_2S + P_2M + P_2L \leq 900$$

$$P_3S + P_3M + P_3L \leq 450$$

Storage:

$$12P_1S + 15P_1M + 20P_1L \leq 13000$$

$$12P_2S + 15P_2M + 20P_2L \leq 12000$$

$$12P_3S + 15P_3M + 20P_3L \leq 5000$$

Demand:

$$P_1S + P_2S + P_3S \leq 750$$

$$P_1M + P_2M + P_3M \leq 1200$$

$$P_1L + P_2L + P_3L \leq 900$$

Non-negativity:

$$\{P_1S, P_1M, P_1L, P_2S, P_2M, P_2L, P_3S, P_3M, P_3L\} \geq 0$$

In short: Determine how many products of each size {S, M, L} to make, in each plant {P₁, P₂, P₃}, per day

Goal: Maximize Profit (in \$)

Limits: Equal percentage of capacity per plant, Capacity (Units / day), Storage (ft² / plant), Demand (units / day)